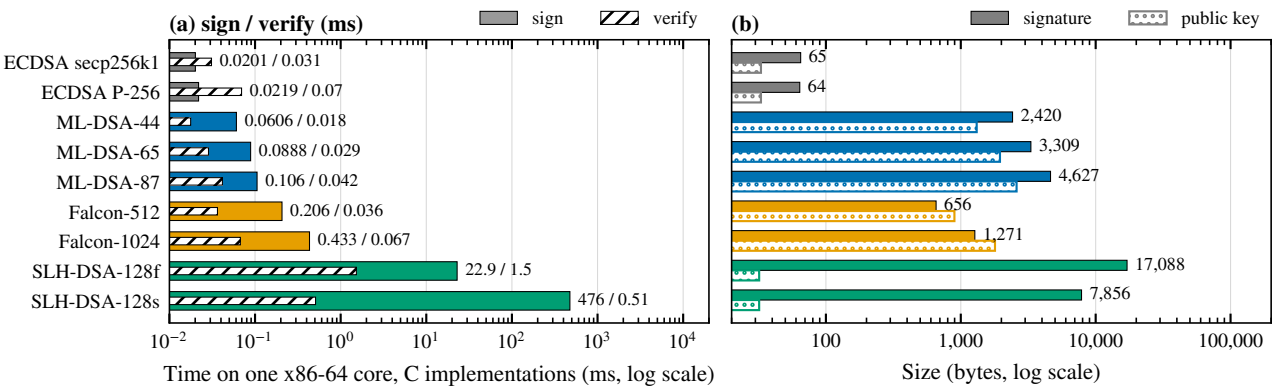

(a) sign / verify (ms)
sign
verify
ECDSA secp256k1
ECDSA P-256
ML-DSA-44
ML-DSA-65
ML-DSA-87
Falcon-512
Falcon-1024
SLH-DSA-128f
SLH-DSA-128s
0.0201 / 0.031
0.0219 / 0.07
0.0606 / 0.018
0.0888 / 0.029
0.106 / 0.042
0.206 / 0.036
0.433 / 0.067
22.9 / 1.5
476 / 0.51
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
Time on one x86-64 core, C implementations (ms, log scale)
(b)
signature
public key
65
64
2,420
3,309
4,627
656
1,271
17,088
7,856
100
1,000
10,000
100,000
Size (bytes, log scale)